

# Cheng Guo

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## CONTACT INFORMATION

Website: <https://chengg04.github.io>

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## EXPERIENCE

**Clemson University**, Clemson, SC December 2021 - present

*School of Mathematical and Statistical Sciences*

Assistant Professor, area: Operations Research (tenure clock start: Fall 2022)

**Columbia University**, New York, NY 2021 - 2022

*Department of Industrial Engineering and Operations Research*

Visiting Researcher

- Host: Daniel Bienstock
- DOE ARPA-E PERFORM (Performance-based Energy Resource Feedback, Optimization, and Risk Management) project

## EDUCATION

**University of Toronto**, Toronto, ON September 2017 - November 2021

*Department of Mechanical and Industrial Engineering*

Ph.D. in Industrial Engineering, GPA: 3.96/4.00

- Advisor: Merve Bodur
- Selected coursework: Stochastic Programming & Robust Optimization, Modeling Interactions on Networks, OM Matching Markets, Mathematical Methods in Power Systems

**Columbia University**, New York, NY 2015 - 2017

*Department of Industrial Engineering and Operations Research*

M.S. in Operations Research

- Selected coursework: Transportation Analytics & Logistics, Optimization I, Programming for Financial Engineering, Seminar on Queueing Theory

**Wuhan University**, Wuhan, China 2011 - 2015

*School of Economics and Management*

B.A. in Economics, B.S. in Mathematics

- Hongyi Honor Program Outstanding Graduates Award
- Selected coursework: Advanced Microeconomics, Industrial Organization, Advanced Financial Theory, Advanced Econometrics, Chaotic Dynamical Systems, Topology

## PUBLICATIONS

(\*: corresponding author; Underline: student coauthor)

6. *Networked Markets with Production and Edge Capacity Constraints: From Competitive Equilibria To Market Entry* [\[full version\]](#)

Cheng Guo\*, Jiayi Wang, Ozan Candogan

**ACM Conference on Economics and Computation (EC)** (2026)

5. *Copositive Duality for Discrete Energy Markets* [\[pdf\]](#)

Cheng Guo\*, Merve Bodur, Joshua A. Taylor

**Management Science** 72.3 (2026): 2022-2040

4. *Tightening Quadratic Convex Relaxations for the Alternating Current Optimal Transmission Switching Problem* [\[pdf\]](#)  
Cheng Guo\*, Harsha Nagarajan, Merve Bodur  
**INFORMS Journal on Computing**, 38.3 (2026): 921-942  
• Appeared as a featured article in the May-June 2026 issue.
3. *Risk-Aware Security-Constrained Unit Commitment: Taming the Curse of Real-Time Volatility and Consumer Exposure* [\[pdf\]](#)  
Daniel Bienstock, Yury Dvorkin, Cheng Guo\*, Robert Mieth, Jiayi Wang  
**IEEE Transactions on Energy Markets, Policy and Regulation** 2.4 (2024): 536-551
2. *Generation Expansion Planning with Revenue Adequacy Constraints* [\[pdf\]](#)  
Cheng Guo\*, Merve Bodur, Dimitri J. Papageorgiou  
**Computers & Operations Research** 142 (2022): 105736
1. *Logic-based Benders Decomposition and Binary Decision Diagram Based Approaches for Stochastic Distributed Operating Room Scheduling* [\[pdf\]](#)  
Cheng Guo\*, Merve Bodur, Dionne M. Aleman, and David R. Urbach  
**INFORMS Journal on Computing** 33.4 (2021): 1551-1569

#### PREPRINTS

3. *Networked Cournot Competition under Congestion* [\[pdf\]](#)  
Cheng Guo\*, Jiayi Wang, Ozan Candogan, submitted  
• Conference version accepted at EC 2026
2. *Pricing Discrete and Nonlinear Markets With Semidefinite Relaxations* [\[pdf\]](#)  
Cheng Guo, Lauren Henderson, Ryan Cory-Wright, Boshi Yang
1. *Incentivizing Investment and Reliability: A Study on Electricity Capacity Markets* [\[pdf\]](#)  
Cheng Guo, Christian Kroer, Yury Dvorkin, Daniel Bienstock

#### HONORS AND AWARDS

- Finalist for student Anna Deza, INFORMS Undergraduate OR Prize Competition, 2020
- Mixed Integer Programming Workshop Student Travel Support, 2019
- Hongyi Outstanding Graduates Award, 2015
- Economics and Management School Scholarship, 2013 - 2014

#### FUNDING

- Clemson University - Clemson Faculty SUCCEEDS: Program 1 (Project Initiation/SEED Funding, awarded to 16 PIs university-wide) (PI), 2024
- University of Toronto - Bert Wasmund Graduate Fellowships in Sustainable Energy Research, 2018

#### TEACHING

##### **Clemson University**

##### *Instructor*

- MATH 8100 - Mathematical Programming (graduate): Fall 2022, Spring 2023, Spring 2024, Fall 2024, Fall 2026
- MATH 4400/6400 - Linear Programming (undergraduate/graduate): Fall 2024, Fall 2025, Spring 2026
- STAT 3090 - Introductory Business Statistics (undergraduate): Spring 2022 (virtual), Fall 2023, Fall 2024, Fall 2025, Fall 2026

## University of Toronto

### *Tutorial Teaching Assistant*

- MIE 562 - Scheduling (undergraduate/graduate): Fall 2019, Fall 2020
- MIE 335 - Algorithms and Numerical Methods (undergraduate): Winter 2019

## Wuhan University

### *Teaching Assistant*

- Probability Theory (undergraduate): Fall 2014

## ADVISING

### **Ph.D. Students**

Lauren Henderson, Benjamin Hamlin (2026, co-advised with Margaret Wiecek)

### **M.S. Students**

Lauren Henderson (2024)

### **Undergraduate Students**

Jiayi Wang (co-advised, Columbia B.S. 2022 → Stanford Ph.D.), Anna Deza (co-advised, U. Toronto B.A.Sc. 2020 → UC Berkeley Ph.D.), Ryan Do (co-advised, U. Toronto B.A.Sc. 2019 → U. Toronto M.Eng.)

### **Ph.D. Thesis Committee Member**

Kristen Joyce, Sarah Kelly (2024)

### **M.S. Thesis Committee Member**

Yunheng Jiang (2022)

## INVITED TALKS

- |                                                       |                |
|-------------------------------------------------------|----------------|
| • Ohio State University, ISE Seminar, Columbus, OH    | October, 2026  |
| • MIT, EIMC2 Seminar, Boston, MA                      | July, 2026     |
| • Cornell University, FIND Seminar, Ithaca, NY        | April, 2024    |
| • Mixed Integer Programming Workshop, Los Angeles, CA | May, 2023      |
| • Polytechnique Montreal, GERAD Seminar, Virtual      | May, 2022      |
| • Discrete Optimization Talks, Virtual                | December, 2020 |

## CONFERENCE PRESENTATIONS

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|---------------------------------------------------------------------------------|----------------|
| • INFORMS Annual Meeting, San Francisco, CA                                     | November, 2026 |
| • INFORMS MSOM Conference, Boston, MA                                           | July, 2026     |
| • Marketplace Innovation Workshop (MIW), Online                                 | May, 2026      |
| • Production and Operations Management Society (POMS) Conference, Reno, NV      | May, 2026      |
| • INFORMS Optimization Society Conference, Atlanta, GA                          | March, 2026    |
| • INFORMS Annual Meeting, Atlanta, GA                                           | October, 2025  |
| • IEEE Power and Energy Society (PES) General Meeting, Austin, TX               | July, 2025     |
| • International Conference on Continuous Optimization (ICCOPT), Los Angeles, CA | July, 2025     |
| • Production and Operations Management Society (POMS) Conference, Atlanta, GA   | May, 2025      |
| • INFORMS Computing Society Conference, Toronto, ON                             | March, 2025    |

- INFORMS Annual Meeting, Seattle, WA October, 2024
- International Symposium on Mathematical Programming, Montreal, QC July, 2024
- INFORMS Optimization Society Conference, Houston, TX March, 2024
- INFORMS Annual Meeting, Phoenix, AZ October, 2023
- INFORMS MSOM Conference, Montreal, QC June, 2023
- INFORMS Annual Meeting, Indianapolis, IN October 2022
- International Conference on Continuous Optimization (ICCOPT), Bethlehem, PA July, 2022
- INFORMS Optimization Society Conference, Greenville, SC March, 2022
- INFORMS Annual Meeting, Virtual October 2021
- International Conference on Game Theory (poster), Virtual July 2021
- IPCO Conference (poster), Virtual June 2021
- CORS Annual Conference, Virtual June 2021
- Mixed Integer Programming Workshop (poster), Virtual May 2021
- Grid Science Winter School (poster), Virtual January 2021
- INFORMS Annual Meeting, Virtual November 2020
- INFORMS Annual Meeting, Seattle, WA October 2019
- DIMACS Workshop on MINLP (poster), Montreal, QC October 2019
- Mixed Integer Programming Workshop (poster), Boston, MA July 2019
- Optimization Days, Montreal, QC May 2019
- INFORMS Computing Society Conference, Knoxville, TN January 2019

#### ACADEMIC SERVICE

- Journal reviewer for *Operations Research*, *Mathematical Programming*, *Management Science*, *Manufacturing & Service Operations Management*, *SIAM Journal on Optimization*, *INFORMS Journal on Computing*, *INFORMS Journal on Data Science*, *Transportation Science*, *Production and Operations Management*, *Computers and Operations Research*, *IEEE Transactions on Power Systems*, *IEEE Transactions on Energy Markets, Policy and Regulation*
- Conference reviewer for *ACM Conference on Economics and Computation (EC)* (2026), *IEEE Power & Energy Society General Meeting (PES)* (2024 - 2026)
- Award committee member for INFORMS ENRE Best Publication Award - Energy (2026), MIP Workshop poster competition (2026)
- Program committee member for Mixed Integer Programming (MIP) Workshop (2026)
- Co-host for the Energy, Market, and Optimization (EMO) Seminar [\[website\]](#)
- Vice President/President-Elect (2026) and President (2027) for INFORMS Junior Faculty Interest Group (JFIG)
- Session chair for INFORMS Optimization Society Conference (2022, 2024, 2026); INFORMS Annual Meeting (2019, 2021, 2022, 2023, 2024, 2026); CORS Annual Meeting (2021)

#### UNIVERSITY SERVICE

- Clemson University SMSS Research Committee (2023-present)
- Co-organizer of University of Toronto MIE UTOrg Seminar (2019-2020)

ADVANCED  
TRAINING

- “Cultivating an Inclusive Classroom Environment”, Academic Impressions

2024

SKILLS

- Programming language: Python, Julia, C++
- Software: Gurobi, CPLEX, Knitro, Mosek

OTHER  
ACTIVITIES

- INFORMS UofT Student Chapter (Honorable Mention, 2020), Vice President (2019-2021)
- Columbia IEOR Mentorship Program, Mentor (2018-2020)
- Wuhan U. Women Soccer Team, Captain (2013-2015)